

Word Problems with Metric Measurements

Lessons

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On the GED math test, you will be expected to know the basic conversions that you have learned in this lesson. If any other measurement fact is needed to solve a problem, the information will be provided with the test item. You can create a unit multiplier from any fact that you are given. A **unit multiplier** is a fraction, written with two different measurement units, that has a value of one.

Example

Hide

Justin needs to know how many inches are equal to 1 meter. He is given the following information: 1 in = 2.54 cm.

Step 1

Justin needs to write a chain of unit multipliers that connects meters and inches. In addition to $1 \text{ in} = 2.54 \text{ cm}$, he needs another fact that connects centimeters and meters. He uses the fact that $100 \text{ cm} = 1 \text{ m}$.

$$\frac{1 \text{ in}}{2.54 \text{ cm}} \times \frac{100 \text{ cm}}{1 \text{ m}}$$

$$1 \cancel{\text{m}} \times \frac{100 \cancel{\text{cm}}}{1 \cancel{\text{m}}} \times \frac{1 \text{ inches}}{2.54 \cancel{\text{cm}}} = 39.37 \text{ inches}$$

Step 2

Multiply. Cancel like measurement units.

$$\frac{1 \text{ in}}{2.54 \text{ cm}} \times \frac{100 \cancel{\text{ cm}}}{1 \text{ m}} = \frac{100 \text{ in}}{2.54 \text{ m}}$$

Step 3

Divide to create a unit rate.

$$100 \div 2.54 \approx 39.37 \text{ inches per meter}$$

There are about **39.37 inches** in 1 meter.

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= Medium

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Category

Lesson 2: The Metric System

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